

Strategy for Charging Optimization:

The following tactics are suggested to maximize charging infrastructure and operations based on the analysis and forecasting of EV charging demand using time-series models and meteorological variables:

1. Management of Peak Demand

Observation: Using historical and forecasted data (from Prophet), peak demand hours were determined.

Approach:

Use time-based pricing to move demand to off-peak hours and discourage billing during peak hours. Make use of scheduling tools that give priority to important or reserved sessions during periods of high demand.

2. Load balancing that responds to weather

Observation: There were correlations found between energy consumption, humidity, and temperature.

Approach:

Pre-allocate energy resources according to weather projections by using dynamic load management. To prevent grid overload on hotter days (when charging demand rises), use smart throttling or staggered charging.

3. Resource Allocation Based on Forecasts

Observation: The hourly forecasts from the prophet model were accurate. Approach:

Pre-plan the availability of chargers and energy purchases using the predicted demand ($\hat{y}$). It is possible to model buffer energy requirements using forecast uncertainty ($\hat{y}_{upper/lower}$).

4. Planning for Infrastructure

Observation: High usage tendencies are consistently observed throughout certain hours.

Approach:

During periods of high demand, increase the deployment of chargers or perform maintenance checks. To meet energy needs during those times, install solar panels or local battery storage.

5. Digital Alerts & User Awareness

Approach:

Notify users via apps about the best times to charge depending on demand and the weather. To aid in user planning, display load curves in real time and in advance at charging stations or in mobile applications.